

MTH212

Examination - January Semester 2026

Statistical Analysis

Thursday, 23 April 2026

04:00 pm - 06:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FOUR (4)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if **FULL** working is shown.
4. This is an **Open-book** examination.

At the end of the examination.

Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.

If you have used more than one answer book, please tie them together with the string provided.

**THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR
SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.**

Answer ALL questions. (Total: 100 marks)

Question 1

A local power supply company selected ten household living in 4-room HDB flat at random and recorded their utilities bill for the last month in Table Q1.

133	119	109	94	127	123	138	147	111	120
-----	-----	-----	----	-----	-----	-----	-----	-----	-----

Table Q1: Monthly Utilities Bills For 4-room Flats

- (a) Describe the data and show estimations for population parameters by calculating the mean, median, variance, standard deviation and range in Table Q1. (5 marks)
- (b) Apply a statistical method to generate a 95% confidence interval for the average bills of 4-room HDB flats using the data given in Table Q1. (5 marks)
- (c) According to Keppel Electric, the average utilities cost for 4-room HDB flat is about 364 kWh costing \$111.57 per month. Interpret the results from the confidence interval in Question 1b and what information can you draw from it. (5 marks)
- (d) Use Python and the data from Table Q1 to implement a bootstrap method. Construct a 95% bootstrap confidence interval for mean utilities cost. Use the `numpy.random.seed(88)` and 5000 iterations. (5 marks)

Question 2

In addition to the utilities bills for 4-room flats, the bills for another ten 5-room flats were randomly selected. Their utilities bills are shown in Table Q2.

4-room flats	133	119	109	94	127	123	138	147	111	120
5-room flats	128	135	140	141	120	115	127	139	132	119

Table Q2: Utilities Bills

- (a) Using any software, apply a suitable hypothesis testing to test if 5-room flat owners are consuming more power at the 0.05 significance level. Conduct the analysis in a step-by-step manner and interpret the results. (15 marks)
- (b) Interpret the results from the hypothesis tests and highlight areas to improve this study. (5 marks)

Question 3

A cement company has developed four different formulations for the production of cement. 4 random samples from each formulation were selected and tested for tensile strength measured in lbs/sq. inch. Table Q3 shows the results of the destructive tests.

Formulation	Tensile Strength (lbs/sq. inch)			
A	3100	3090	3220	3150
B	2620	2780	2675	2750
C	2855	2850	2985	3050
D	2950	2755	2815	2965

Table Q3: Tensile Strength

- (a) Apply a suitable hypothesis testing to test if there is different in tensile strength between formulation. Conduct the analysis in a step-by-step manner and draw your conclusions. (15 marks)
- (b) Interpret the results from the hypothesis tests and highlight any additional analysis needed. (5 marks)

Question 4

A university is investigating whether there is an association between gender and the preferred study method among students. Students were asked if they prefer Group Study, Individual Study or Online Study. A random sample of 160 students was surveyed, and the results are shown Table Q4:

Gender	Group	Individual	Online
Males	20	35	30
Females	30	25	20

Table Q4: Observed Values

- (a) Apply a suitable hypothesis testing to test if there is a difference in study method between genders. Conduct the analysis in a step-by-step manner and draw your conclusions. (15 marks)
- (b) Interpret the results from the hypothesis tests. (5 marks)

Question 5

A university is studying the factors that influence students' final exam scores (y). Three predictors are considered: x_1 : Number of study hours per week. x_2 : Number of hours of sleep per night. x_3 : Stress level (measured on 1-10 scales, with higher values indicating more stress). The data is shown in Table Q5.

S/No	Exam Score y	Study Hours x_1	Sleep Hours x_2	Stress x_3
1	68	12	6	7
2	70	18	5	8
3	82	22	7	6
4	90	28	8	5
5	66	16	6	9
6	85	25	7	6
7	72	20	5	7
8	92	30	7	5
9	69	14	6	8
10	88	26	8	5

Table Q5: Exam Scores

- (a) Calculate the adjusted coefficient of correlation R^2 and interpret its meaning. What is the difference between R^2 and adjusted R^2 . Round all numbers to **TWO (2)** decimals. (5 marks)
- (b) Implement linear regression models to analyze the data in Table Q5. Determine the regression equation and interpret the coefficients. (5 marks)
- (c) Using any software, apply a suitable hypothesis testing to test if the regression model is significant and whether coefficients are all zero. Conduct the analysis in a step-by-step manner and draw your conclusions. (10 marks)

----- END OF PAPER -----